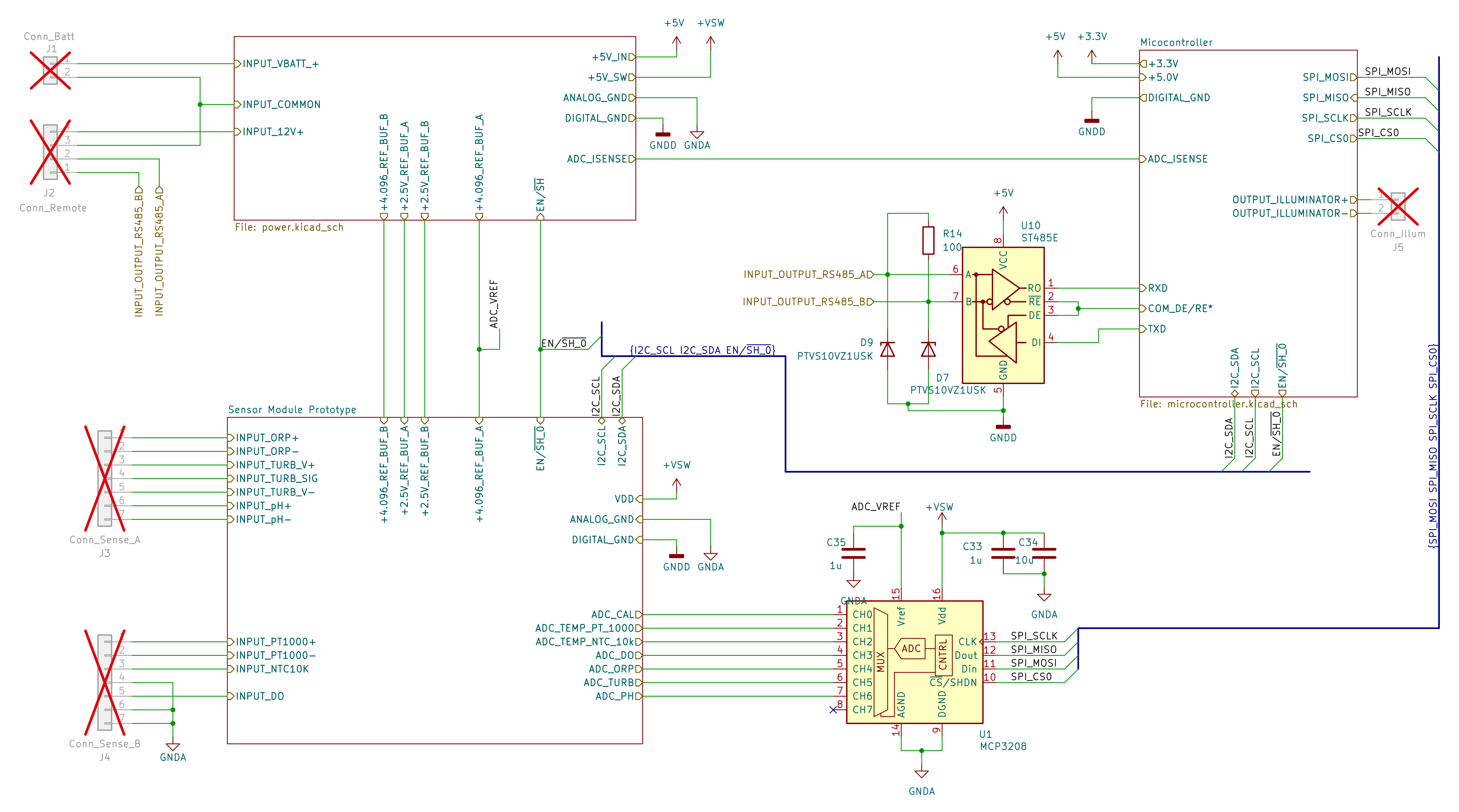

Using Cat5 cable, the characteristic impedance is 100 Ohm. Adding a DC termination resistor of 100 Ohm to both ends of the cable should suffice for termination. This adds a DC load to the drivers though and you should consider designing an AC termination instead (Rt in series with a cap -- this messes up the rise and fall times though so you have to do it carefully).

ADC_VREF

PWR_FLAG

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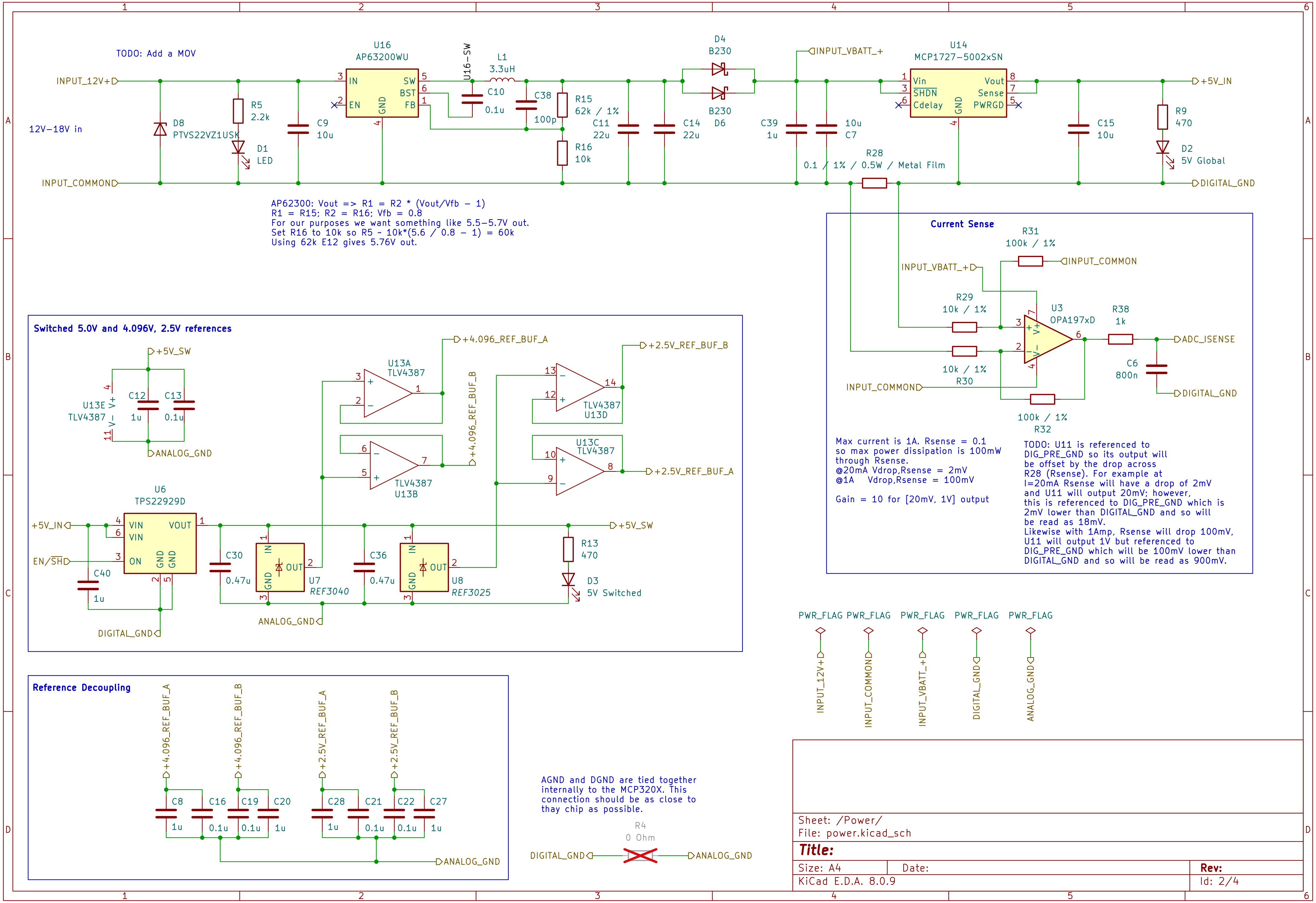
Size: A4 | Date: | **Rev:**

KiCad E.D.A. 8.0.9 | Id: 1/4


TODO: Add a MOV
12V−18V in
INPUT_12V+
INPUT_COMMON
D8
PTVS22VZ1USK
R5
2.2k
D1
LED
C9
10u
U16
AP63200WU
IN
EN
GND
SW
BST
FB
U16−SW
L1
3.3uH
C10
0.1u
C38
100p
R15
62k / 1%
R16
10k
C11
22u
C14
22u
D4
B230
B230
D6
INPUT_VBATT_+
C39
1u
10u
C7
U14
MCP1727−5002xSN
Vin
SHDN
Cdelay
GND
Vout
Sense
PWRGD
C15
10u
R9
470
D2
5V Global
+5V_IN
DIGITAL_GND
R28
0.1 / 1% / 0.5W / Metal Film
AP62300: Vout => R1 = R2 * (Vout/Vfb − 1)
R1 = R15; R2 = R16; Vfb = 0.8
For our purposes we want something like 5.5−5.7V out.
Set R16 to 10k so R5 ~ 10k*(5.6 / 0.8 − 1) = 60k
Using 62k E12 gives 5.76V out.
Current Sense
R31
100k / 1%
INPUT_VBATT_+
INPUT_COMMON
R29
10k / 1%
U3
OPA197xD
R38
1k
ADC_ISENSE
C6
800n
DIGITAL_GND
10k / 1%
R30
INPUT_COMMON
100k / 1%
R32
Max current is 1A. Rsense = 0.1
so max power dissipation is 100mW
through Rsense.
@20mA Vdrop,Rsense = 2mV
@1A Vdrop,Rsense = 100mV
Gain = 10 for [20mV, 1V] output
TODO: U11 is referenced to
DIG_PRE_GND so its output will
be offset by the drop across
R28 (Rsense). For example at
I=20mA Rsense will have a drop of 2mV
and U11 will output 20mV; however,
this is referenced to DIG_PRE_GND which is
2mV lower than DIGITAL_GND and so will
be read as 18mV.
Likewise with 1Amp, Rsense will drop 100mV,
U11 will output 1V but referenced to
DIG_PRE_GND which will be 100mV lower than
DIGITAL_GND and so will be read as 900mV.
Switched 5.0V and 4.096V, 2.5V references
+5V_SW
U13E
TLV4387
C12
1u
C13
0.1u
ANALOG_GND
U13A
TLV4387
+4.096_REF_BUF_A
+4.096_REF_BUF_B
U13B
TLV4387
U13D
TLV4387
+2.5V_REF_BUF_B
U13C
TLV4387
+2.5V_REF_BUF_A
U6
TPS22929D
VIN
VIN
ON
GND
GND
VOUT
+5V_IN
EN/SHD
C40
1u
DIGITAL_GND
C30
0.47u
U7
REF3040
IN
OUT
GND
ANALOG_GND
C36
0.47u
U8
REF3025
+5V_SW
R13
470
D3
5V Switched
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
INPUT_12V+
INPUT_COMMON
INPUT_VBATT_+
DIGITAL_GND
ANALOG_GND
Reference Decoupling
+4.096_REF_BUF_A
+4.096_REF_BUF_B
+2.5V_REF_BUF_A
+2.5V_REF_BUF_B
C8
1u
C16
0.1u
C19
0.1u
C20
1u
C28
1u
C21
0.1u
C22
0.1u
C27
1u
ANALOG_GND
AGND and DGND are tied together
internally to the MCP320X. This
connection should be as close to
thay chip as possible.
R4
0 Ohm
DIGITAL_GND
ANALOG_GND
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Size: A4
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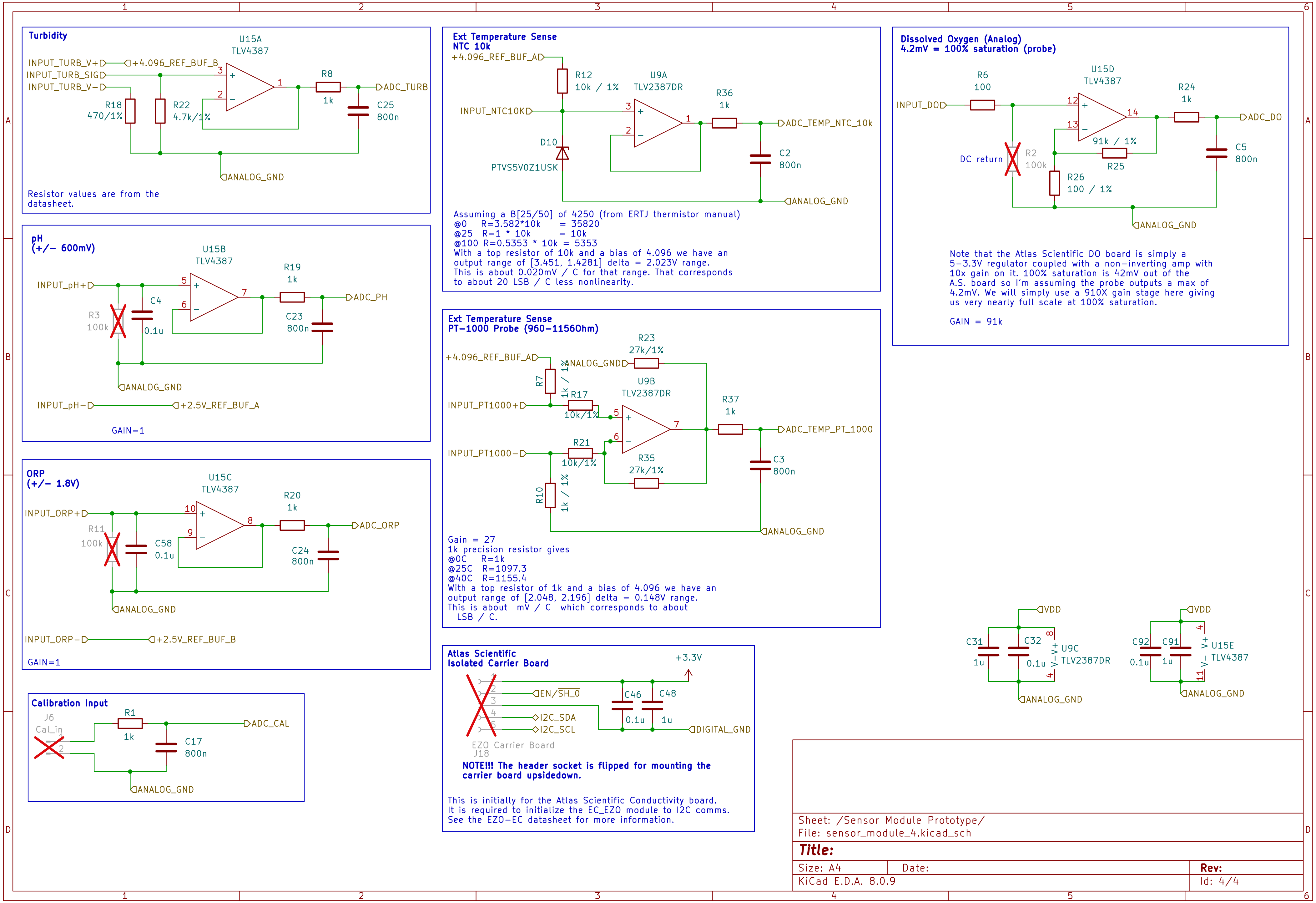
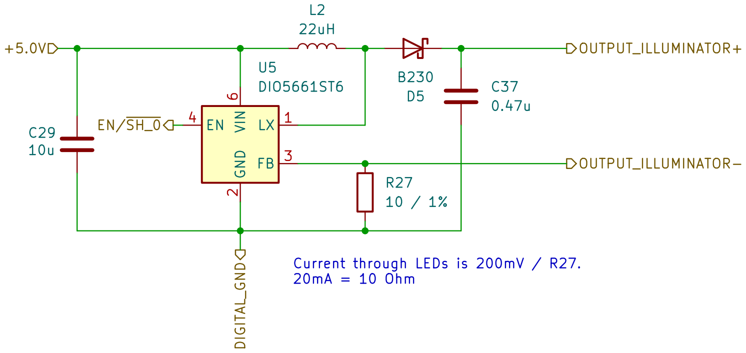
Rev:
KiCad E.D.A. 8.0.9
Id: 2/4

Amphenol TSW-10 A washing machine sensor
but it's calibrated!



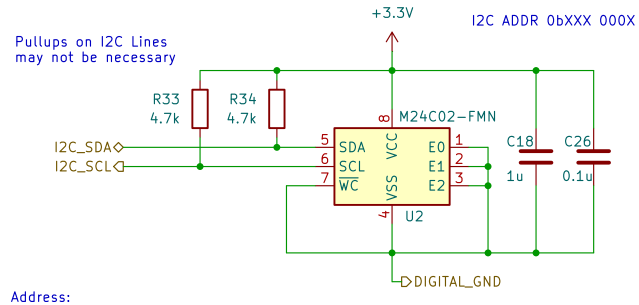
We need a powered output and switch for the LED ring. Ideally, a constant-current LED driver would be used although I'd like to keep the option open to use a multi-spectral led source which will probably want one channel per LED. For now, maybe just a constant-current serial output?



Current through LEDs is 200mV / R27.
20mA = 10 Ohm

Pullups on I2C Lines
may not be necessary

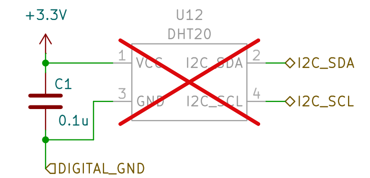
I2C ADDR 0bXXX 000X

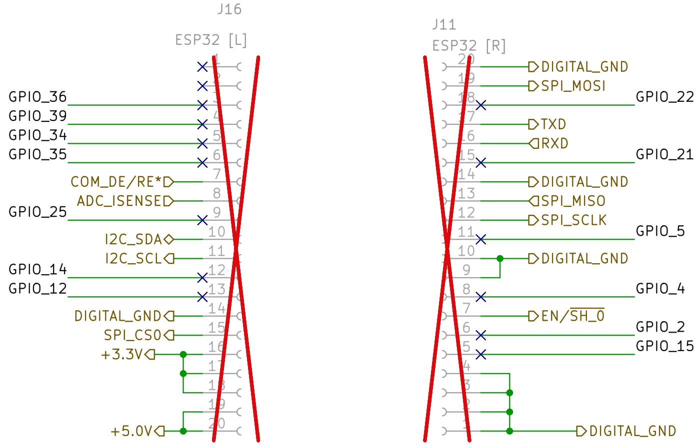


Address:
0xA0 [0b1010 0(p3) 0(p2) 0(p1) R/*W]

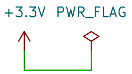
Internal Temp / Humidity (DTH 20)

I2C Addr 0x38





ESP32-Freenove



NOTE: The ESP32 carrier board has headers on the bottom.
Thus the carrier board header socket pinout is identical to the
ESP32 board as viewed from above.

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Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 6/4